

1. Foundations of AI and Machine Learning

- 1.1 ML vs Traditional Programming
- 1.2 Difference between AI, ML, and DL
- 1.3 Types of Learning
- 1.4 Supervised Learning
- 1.5 Unsupervised Learning
- 1.6 Semi-Supervised Learning
- 1.7 Reinforcement Learning

2. Types of Machine Learning Tasks

- 2.1 Classification
- 2.2 Regression
- 2.3 Clustering

3. Data Preprocessing

- 3.1 Handling Missing Values
- 3.2 Feature Scaling
- 3.3 Data Encoding Methods

4. Linear Regression

- 4.1 Linear Regression
- 4.2 Linear Regression Model Formulation
- 4.3 Linear Regression Assumptions
- 4.4 Limitations of Linear Regression

5. Logistic Regression

- 5.1 Logistic Regression
- 5.2 Logistic Regression Model Formulation
- 5.3 Sigmoid Function
- 5.4 Logistic Regression Assumptions
- 5.5 Limitations of Logistic Regression

6. Evaluation Metrics

- 6.1 Classification Metrics
 - 6.1.1 Confusion Matrix
 - 6.1.2 Precision and Recall
 - 6.1.3 F1 Score
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 - 6.2.1 Mean Absolute Error (MAE)
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7. Decision Trees

7.1 Introduction to Decision Trees

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8.2 Multi Layer Perceptron

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9.4 One Hot Encoding

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